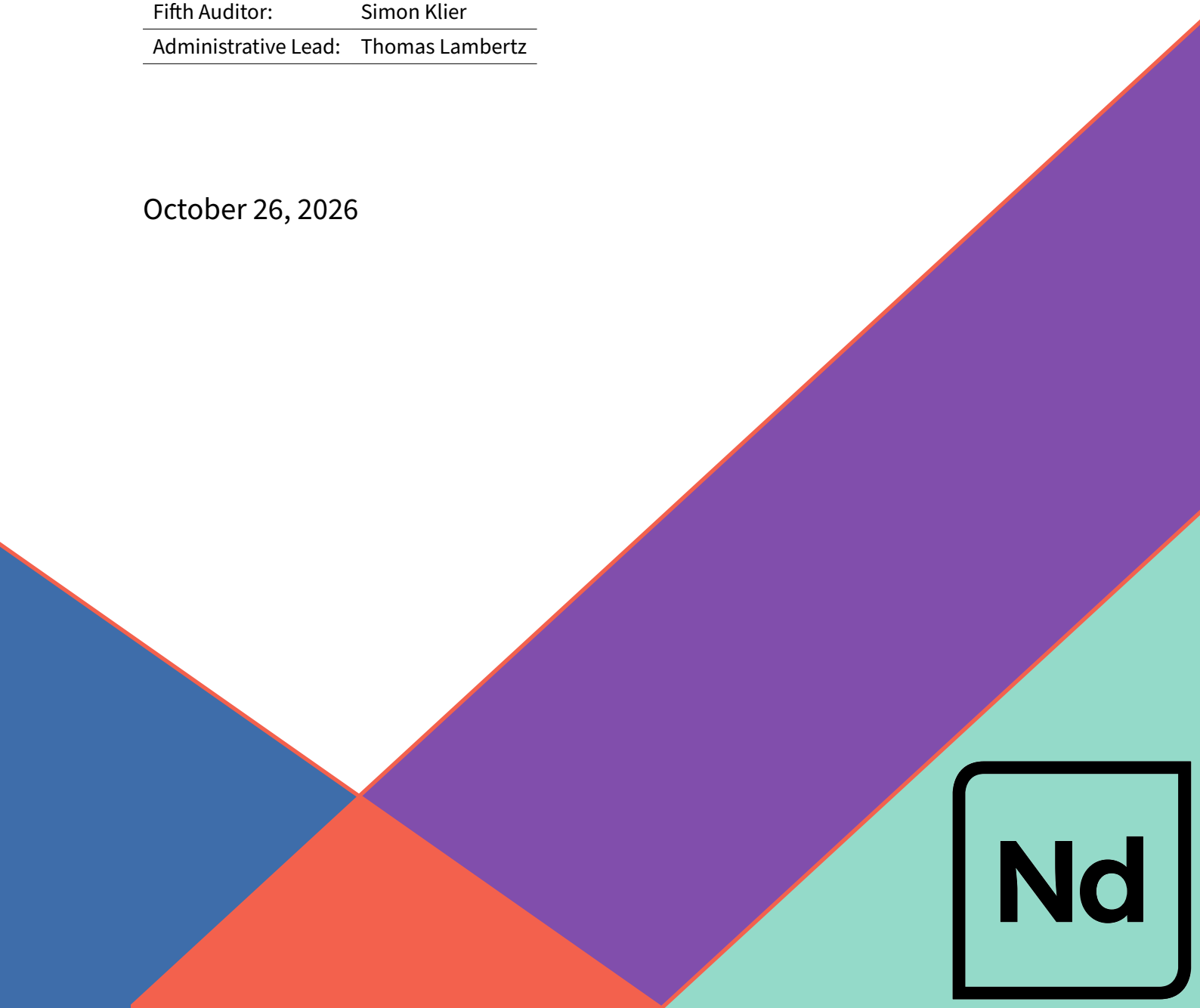
Point Release Audit - Agave v4.4

conducted by Neodyme AG

Lead Auditor:	Thomas Lambertz
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October 26, 2026



1 | Executive Summary

On October 26, 2026, Neodyme completed a Point Release Audit of Anza's Agave changes between the `v4.3.0` tag (`825efd1829`) and the `v4.4.0-rc.1` tag (`b6149a719f`). The review covered six assigned functional areas, including the v4.4 changes to Alpenglow.

Neodyme identified two issues during this audit. Both were already known to Anza and are therefore classified as informational in this report. No critical- or high-severity issues were identified.

2 | Scope

Reviewed Revision

Boundary	Reference	Commit
Start	`v4.3.0`	`825efd1829`
End	`v4.4.0-rc.1`	`b6149a719f`

Neodyme reviewed changes to:

- **Transaction Ingestion Pipeline:** transaction-version gate cleanup, QUIC stream throttling and memory limits, UDP packet handling, and transaction forwarding.
- **Leader Schedule & Block Production:** bounded PoH-to-broadcast channels, block-component parsing, leader handover, and the Alpenglow block-production path.
- **Block Propagation (Turbine/Shreds/XDP):** shred proof and layout validation, Turbine broadcast and repair, and blockstore handling of equivocation and UpdateParent slots.
- **Accounts DB & State Storage:** account-index ref-count removal, account cleaning and packing, the new split-file format, snapshot encoding, and startup hints and hashing.
- **Gossip & P2P Communication:** wincode-only message decoding, duplicate-shred proof validation, network buffering, and the externalized Geyser plugin interface.
- **Alpenglow Consensus Migration:** BLS vote batching and aggregate verification, genesis-vote admission, clock and progress tracking, certificate transport, and fast-leader-handover paths.

Exclusions

The audit was limited to the six functional areas listed above. Separately audited external cryptography components and non-production components such as CI, scripts, CLI, RPC, tests, and benchmarks were outside scope. Cross-flow changes remained in scope where they affected the reviewed areas.

4 | About Neodyme

Security is difficult.

To understand and break complex things, you need a certain type of people: people who thrive in complexity, who love to play around with code, and who do not stop exploring until they fully understand every aspect of it. That's us.

Our team never outsources audits. Having found over 80 high- or critical-severity bugs in Solana's core code itself, we believe that Neodyme hosts the most qualified auditors for Solana programs. We have also found and disclosed critical vulnerabilities in many of Solana's top projects and responsibly disclosed issues that could have resulted in the theft of over $10B in TVL on the Solana blockchain.

All of our team members have a background in competitive hacking. During hacking competitions known as CTFs, we competed and collaborated while finding vulnerabilities, breaking encryption, reverse engineering complex algorithms, and much more. Many of our team members have won national and international competitions and continue to rank highly in some of the world's hardest CTF events.

In 2020, some of our members started experimenting with validators and became active members of the early Solana community. After reporting several bugs, the Solana Foundation contracted us for source-code auditing in 2021. As a result, Neodyme was born.

